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SUPERSEDING
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MILITARY SPECIFICATION

BOOTS, TOXICOLOGICAL AGENTS PROTECTIVE, M2A1

* This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers butyl rubber boots, with steel safety toes, designed for wear over stockings.

1.2 Classification. The boots shall be of the following types and sizes as specified (see 6.2).

Type I - Knee length
Type II - Short length (with leg closure ring)

Schedule of sizes

5 6 7 8 9 10 11 12 13 14 15

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8430

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SPECIFICATIONS

FEDERAL

A-A-203	-	Paper, Kraft, Untreated
QQ-A-250/11	-	Aluminum Alloy 6061, Plate and Sheet
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-A-8625	-	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-E-20652/1-		Eyelets, Metallic, Rolled Flange Type; and Eyelet Washer
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-595	-	Colors

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-282	-	Filter Units, Protective Clothing, Gas-Mask Components and Related Products, Performance Test Methods
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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DRAWING

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

8-1-182 - Leg and Boot Closure Ring

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 412 - Rubber Properties in Tension
D 1630 - Rubber Property - Abrasion Resistance (NBS Abrader)
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

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3.2 Material (see 6.5).

3.2.1 Rubber. The compounds to be used in assembly of the boot (except the cement and the stiffening counter) shall be made from butyl rubber and shall not contain natural, synthetic (other than butyl rubber), or reclaimed rubber. Reclaimed butyl rubber in amounts not to exceed 5 percent of the total batch weight may be used in compounding the outsole. Use of rag stock in natural rubber or synthetic rubber compound is allowed in the stiffening counter as referenced in table III. The rubber compounds shall meet the requirements in table I when test as specified in 4.4.1.

TABLE I. Requirements for rubber compounds

Characteristics	Upper compound	Outsole and heel compound
Tensile strength, minimum:		
Initial (psi)	1000	-
After accelerated aging, percent of initial strength	75	
Ultimate elongation, percent, minimum	400	-
Abrasive index, minimum	-	50

3.2.2 Cotton fabrics. The cotton fabrics to be coated shall have a pH of not less than 5.5.

3.2.3 Eyelet (for type I). The eyelet shall conform to dash No. ABE 129 of MIL-E-20652/1. As an alternate, an uncoated Junior eyelet and washer with an inside diameter of 1/4 inch may be used.

3.2.4 Adhesives. Only neoprene or butyl rubber adhesive is authorized for seams in forming the boot on the last and to supplement natural tack of unsupported compound components laid one over the other. The adhesive shall be compounded to have an optimum cure duplicating that of the butyl rubber compound. The adhesion of these compounds shall be no less than 6 pounds when tested as specified in 4.4.1.

3.2.5 Steel box toe. The steel box toe shall conform to the toe of the last and shall be fabricated from cold-rolled carbon steel. After heat treatment, the steel box toe shall conform to the requirements of table II. The steel box toe shall be thoroughly cleaned and be given a zinc compound coating, parkerizing, or equivalent.

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TABLE II. Physical requirements, steel box toe

Hardness (Rockwell C)	Carbon content, percent	Thickness inch	Trade pattern
42-52	0.50 to 0.82	0.050 to 0.058	931 or equal

3.2.6 Leg closure ring (for type II). The leg closure ring shall be made in accordance with Drawing 8-1-182. The closure ring shall be made of aluminum alloy conforming to temper T6 of QQ-A-250/11 and anodized in accordance with type II, class 2 of MIL-A-8625. Only the male portion of the ring shall be incorporated in the boot.

3.3 Construction.

3.3.1 Lasts. Each manufacturer shall make lasts which shall have the following dimensions:

3.3.1.1 Ankle girth. The circumference of the ankle of the boot lasts for both type I and type II shall be not less than 11 inches for size 9 with a grading of 1/4 inch between sizes.

3.3.1.2 Top girth.

3.3.1.2.1 Type I. The circumference measurement of the outside top of the boot last shall be not less than 15-5/8 inches for size 9 with a grading of 1/4 inch between sizes. The measurement shall be made at a point measuring up 16 inches from center back of heel area.

3.3.1.2.2 Type II. The diameter of the top of the boot last shall be such that when the boot is removed from the last and the top is folded over, the inside diameter of the boot top measured through the center of the ring, shall be as shown on figure 1.

3.3.2 Upper compound components. The following components of the boots shall be made from upper compound in the thickness specified:

Component	Overall thickness, inches minimum
Ankle piece	0.040
Heel piece	0.040
Leg cover	0.040 1/
Backstay	0.040
Inner vamp	0.040
Outer vamp	0.040
Gum toe cap	0.030
Foxing	0.100 2/

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Inside foxing (optional)	0.030 3/
First binder sole (optional)	0.030
Second binder sole (optional)	0.030
Welt	0.160 4/
Binding	0.040 5/

1/ The leg cover shall be made from at least three plies of upper compound.

2/ May have feathered or skived edges. When inside foxing is used, the thickness of the foxing (outside) may be a minimum of 0.080.

3/ Shall be upper compound, or 5.4 ounce minimum woven cloth as specified in table III.

4/ Shall be not less than 5/16 inch in width and the ends shall taper to a feathered edge. If the outsole is designed to physically incorporate welt dimensions within itself, a separate welt is not required.

5/ Type I shall be 7/16 ± 1/16 inch wide and type II shall be 15/16 ± 1/16 inch wide.

3.3.3 Cloth base components. The construction of the components shall be as specified in table III.

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TABLE III. Requirements of components using cotton cloth

Component	Weight of cloth (oz./sq. yd.) minimum Woven	Knitted	Frictioned Both sides	One side	Calender coated	Spread and calender coated	Coated	Overall thickness (inches) minimum	Additional construction details
Leg lining and toe lining	--	7.0	--	--	--	On one side with no less than 0.008 inch of lining compound	--	--	--
Friction heel piece Friction back strip Friction inner vamp	--	5.4	X	--	On one side with lining compound	--	--	0.025	--
Inside foxing (optional)	5.4	--	--	--	X	On one side with upper compound	--	--	0.006
Stiffening counter	5.4	--	X	--	On one side with upper compound	--	--	0.040	Rag stock in natural or synthetic rubber may be used as an alternate for upper compound
Filling sole	5.4	--	X	--	--	--	On one side with stiffening compound	0.55	--

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TABLE III. Requirements of components using cotton cloth (cont'd)

Component	Weight of cloth (oz./sq. yd.) minimum Woven	Knitted	Frictioned Both sides	One side	Calender coated	Spread and calender coated	Coated	Overall thickness (inches) minimum	Additional construction details
Skive sole	5.4	--	X	--	--	--	On one side with stiffening compound	0.55	--
Insole (1)	5.4	--	--	X	--	--	On frictioned side with stiffening compound	0.115	Both insoles (1) and (2) shall be passed through a calender to assemble the coated side of insole (1) to one side of insole (2). The overall thickness of the assembly shall be not less than 0.140 inch
Insole (2)	5.4	--	--	X	--	--	On one side with lining compound, away, from insole (1)	--	--

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3.3.4 Outsole and heel. The outsole and heel shall be made from outsole and heel compound and shall be molded in the outsole mold. The heel shall be an integral part of the outsole. The outsole and heel shall carry an open channel or hobnail tread design to facilitate cleaning or decontamination after use. The minimum thickness for the outsole and heel shall be as specified on figure 1.

* 3.3.5 Gum toe cap. An overlay of yellow compound no less than 0.030 inch thick shall be applied to the outer surface of the boot toe. The color of the overlay shall be any of the following colors in FED-STD-595: 13432, 13538, 13655, 23538, 23655 and 33538. This ply of unsupported gum shall be applied so that it covers the main part or full coverage of the metal safety cap and extends down to or under the foxing.

3.3.6 Assembly of boot to closure ring (type II). The boot shall be bonded to the male portion of the closure ring with butyl rubber specified in 3.2.1 and the adhesive specified in 3.2.4. The ring shall be installed as indicated below and as shown on figure 2. The attached ring shall show no slip or separation from the boot when tested as specified in 4.4.5.

a. Coat inside leg lining with adhesive from top to bottom of binding strip.
b. Apply uncured butyl strip a minimum of 0.040 inch thick to the inside top of the leg lining and fold top of leg inward.
c. Solvent clean the ring.
d. Apply adhesive to the ring.
e. Wrap one layer of uncured butyl stock a minimum of 0.040 inch thick around the bonding area on the outside circumference of the ring.
f. Solvent wipe folded down part of boot and allow to dry.
g. Apply wet coat of adhesive to folded part of boot and insert the ring (male disconnect).

3.4 Height.

* 3.4.1 Type I height. The inside height of the boot shall be 16 $\pm$ 1/2 inches from the top of the insole to the top of the boot. The overall height of the right and left boots comprising a pair shall not vary by more than 1/4 inch.

* 3.4.2 Type II height. The inside height of the boot shall be 11 $\pm$ 1/2 inches from the top of the insole to the top edge of the binding. Before assembling the closure ring to the boot, a minimum of 1 inch of the rubber leg portion must extend beyond the top of the binding. This is then folded down over the butyl strip incorporated during attachment of the closure ring and cannot be accurately measured after attachment of ring. The overall height of the right and left boots comprising a pair shall not vary by more than 1/4 inch.

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3.5 Finish and color. The boots shall have a natural heater finish and the color shall be black or dark green except that the toe cover shall be yellow (see 3.3.5).

3.6 Vulcanization. The boots shall be vulcanized under pressure to insure a watertight, compact unit. The boots may be dusted with cornstarch before curing.

3.7 Performance.

3.7.1 Leakage. After vulcanization, the boot shall be tested for leaks by the contractor in accordance with the test specified in 4.5.7.

3.7.2 Toxicological agents resistance.

3.7.2.1 Mustard vapor penetration. The mustard vapor penetration time of specimens cut from sample boots shall be at least 8 hours when tested as specified in 4.4.5.

3.7.2.2 GB vapor penetration. The GB penetration time of specimens cut from sample boots shall be at least 8 hours when tested as specified in 4.4.5.

3.7.3 Low temperature fracture. The boot shall not fracture at minus 25 ± 2°F when tested as specified in 4.4.5.

3.7.4 Leg lining adhesion. The adhesion of the leg lining to the gum leg cover shall be not less than 5 pounds at a pull rate of 2 inches per minute when tested as specified in 4.4.5.

3.7.5 Replacement of defective components. During the cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.3 shall be removed from production and replaced with nondefective and properly matched components.

3.8 Marking.

3.8.1 Manufacturer's name. At the contractor's option, a label bearing the manufacturer's name or trademark shall be located as specified on figure 1.

3.8.2 Size. The size shall be legibly stamped on the outside at the arch with red or aluminum indelible ink on raised figures which shall be at least 3/8 inch high. As an alternative method, the size may be branded or engraved on the breast (forward vertical face) of the heel.

3.8.3 Interior marking. The following information shall be stenciled or stamped below the binding in the interior, on either side of each boot: Nomenclature of item, lot number, size, contract number, national stock number, and manufacturer.

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3.9 Workmanship. The finished boots shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested for the characteristics specified in table VI. The presence of any defect or failure of any test shall be cause for rejection of the first article.

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4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.4.1.1 Component testing. In addition, testing shall be performed on components and materials listed in table IV. For the adhesive, the lot size shall be expressed in gallons. The sample unit for testing shall be a representative sample of the adhesive. For the upper and outsole compounds, the lot size shall consist of the amount of rubber compound that is produced or compounded at any one time. The composite samples of rubber shall be vulcanized in the same manner employed in the manufacture of the boots. The sample unit for upper compound tests shall be two pieces of upper compound 6 by 6 inches and 0.040 ± 0.010 inch thick. The sample unit for abrasive index test shall be one piece of outsole compound 6 by 6 inches and 0.125 ± 0.010 inch thick. All test reports shall contain the individual values used in expressing the final results. The lot shall be unacceptable if the representative sample of adhesive or any of the composite samples of rubber fail to meet any of the test requirements specified. The lot sizes and sample sizes shall be in accordance witht the following:

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE IV. Component tests

Component	Characteristic	Requirement paragraph	Test method
Upper compound	Tensile strength		
	- Initial	3.2.1	4.5.3
	- After aging	3.2.1	4.5.3
	Ultimate elongation	3.2.1	4.5.3
Outsole and heel compound	Abrasive index	3.2.1	4.5.4
Boot adhesive	Adhesion	3.2.4	4.5.1

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4.4.1.2 Certification. A certificate of compliance shall be furnished with each shipment or lot stating the following:

a. The boots are made from butyl rubber and contain no natural, synthetic other than butyl rubber, or reclaimed rubber; except as specified for the outsole or counter in 3.2.1. The contractor shall furnish a certificate of compliance stating that the leg cover is made of at least three plies of upper compound.

b. The type and weight of each cotton cloth and where each was used (see table III).

c. That all base cloths used have a pH not less than 5.5.

d. That the steel top cap conforms to the requirements in 3.2.5 and table II.

e. That the leg closure ring meets the requirements of 3.2.6 (type II only).

f. That the boot adhesive is either neoprene or a butyl base type and has a curing time duplicating that of the butyl rubber.

g. That the dimensions for lasts conform to the requirements of 3.3.1.1, 3.3.1.2.1, and 3.3.1.2.2.

h. That the leg closure ring is bonded with butyl rubber and adhesive specified in 3.3.6 (type II only).

i. That the thickness of the upper compound components conforms to the requirements of 3.3.2.

j. That the thickness of the outsole and heel conforms to the requirements of figure 1.

k. That a minimum of 1 inch of the rubber leg portion extends beyond the top of the binding in accordance with 3.4.2 (type II only).

* 4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing operation to determine whether construction details which cannot be examined in the finished product are in accordance with specified requirements. Whenever nonconformance is noted, corrections shall be made to the items affected and lot in-process. Items which cannot be corrected shall be removed from production.

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* 4.4.2.1 Critical defect examination of boots. Prior to performing the end item sampling examination required in 4.4.3, the boots shall be 100 percent examined for the critical defects in table V. This 100 percent critical defect examination shall become a part of the contractor's inspection system or quality program. Any boot found to contain a critical defect shall be rejected.

* 4.4.3 End item visual examination. The boots shall be examined for the defects listed in table V. The lot size shall be expressed in units of boots. The sample unit shall be one boot. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects. The finding of any critical defect shall be cause for rejection of the lot.

TABLE V. End item visual defects

Defect	Critical	Major	Minor
Any cut, tear, or hole in upper material	X		
Any cut, tear, or hole through soling material	X		
Any repair or patch	X		
Any component damaged or misplaced, or any operation omitted or not properly performed which may result in injury to the user	X		
Poor or uneven lasting, seriously affecting serviceability or appearance.		X	
Poor or uneven lasting, affecting serviceability or appearance but not seriously.			X
Any component missing.		X	
Air pocket or channel under 1/4 square inch in area in any seam.			X
Air pocket or channel 1/4 square inch or more in area in any seam.		X	
Marking (contractural, size, etc.) missing, incomplete, incorrect, not applied in the specified manner, misplaced, or illegible.			X
Any pit over 1/8 inch in any lateral dimension.		X	
Any pit penetrating through the outer layer of rubber and into the second layer of rubber.		X	
Pits more than 1/16 inch but not more than 1/8 inch in any lateral dimension not penetrating the outer layer of rubber and in sufficient number to average more than one pit per square inch over the entire upper area.			X
Boots not properly mated, i.e., not right and left of same size		X	
More than 1/4 inch variation between right and left boot comprising a pair			X

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TABLE V. End item visual defects (cont'd)

Defect	Critical	Major	Minor
Color of toe overlay not as specified.		X	
Finish of boot not as specified		X	
Type not as specified.		X	

4.4.4 End item dimensional examination. The boots shall be examined for the dimensions specified in paragraphs 3.4.1 and 3.4.2 for the inside height requirements specified. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of one boot. The sample unit shall be one boot and the selection shall be by pairs. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing. The boots shall be tested for the characteristics listed in table VI. For closure ring adhesion test, the lot size shall be expressed in units of one boot, the sample unit shall be one boot, and the inspection level shall be S-1. For leakage test, the lot size shall be expressed in units of one boot, the sample unit shall be one boot, and the inspection level shall be S-4. For the other tests, fifteen boots shall be selected from the lot (ten for the toxicological agents resistance tests and five for the low temperature fracture and leg lining adhesion tests). Any test failure shall be cause for rejection of the lot.

TABLE VI. End item tests

Characteristic	Requirement paragraph	Test method
Closure ring adhesion	3.3.6	4.5.2
Leakage	3.7.1	4.5.7
Toxicological agents resistance 1/	3.7.2	4.5.8
Low temperature fracture	3.7.3	4.5.5
Leg lining adhesion	3.7.4	4.5.6

1/ The toxicological resistance tests for mustard vapor penetration and GB vapor penetration shall be performed by the U.S. Army Armament, Munitions and Chemical Command (see 6.6). The test specimens shall be cut from the boots at the test facility.

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* 4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container.
Content	Number of boots per shipping container is more or less than specified.

* 4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

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4.5 Methods of inspection.

4.5.1 Adhesion test. The samples for the adhesion test shall be prepared as follows: Warm a laboratory mill load of upper compound. Tighten rolls to a setting which will yield a sheet no more than 0.035-inch thick after slab-off and cooling. Cut two 6 by 6-inch sections from the smoothest part of the sheet. Apply one coat of adhesive to one surface of each section and allow the adhesive to dry until firm to the touch. Lay a 2 by 6-inch strip of Holland cloth along one edge of one section. Pair the sections so that the pair will include the Holland cloth strip. Assemble the pair starting with the edges opposite the Holland cloth strip, utilizing a hand roller to minimize air entrapment. Cure the assembly in the same manner as that planned for the finished boots and age 24 hours at 212°F. From the cured assembly, cut six 1.0-inch wide test specimens, perpendicular to the Holland strip and running the full length of the section. Remove the Holland cloth to separate the ends. The separated ends shall then be inserted in the clamps of the tester and the adhesion determined in accordance with Method 5950 of FED-STD-191, except that the pull rate shall be 2 inches per minute.

4.5.2 Closure ring adhesion test (type II). The boot to be tested shall be hung upright by having the closure ring clamped in a horizontal position. The clamping shall be done symmetrically at four points around the periphery of the ring. A 25-pound weight shall be placed in the heel area inside the boot to subject the adhesive bond between the ring and boot to a constant weight. The weight shall remain in the boot for 4 hours after which it shall be removed and the seal examined. Any slippage or separation of the closure ring from the boot shall be considered a test failure. The 25 pounds of weight shall have a horizontal cross-section of approximately 3-1/2 by 4-1/2 inches. Note: A set of five lead weights each weighing 5 pounds has been used satisfactorily. The results shall be reported as pass or fail.

4.5.3 Tensile strength and ultimate elongation tests. Ten dumbell specimens as specified in ASTM D 412 (Method A) shall be cut from the test sample using Die C. Five of these specimens shall be aged at 158° ± 2°F for 7 days prior to testing for tensile strength. All ten specimens shall be tested in accordance with ASTM D 412 (Method A). The five unaged specimens shall also be tested for ultimate elongation in accordance with ASTM D 412 (Method A).

4.5.4 Abrasive index test. Testing for index of abrasion shall be performed in accordance with ASTM D 1630.

4.5.5 Low temperature fracture test. Expose the five sample boots and a 1.00 ± 0.05 pound steel ball for 24 continuous hours to a temperature of minus 25° ± 2°F. Support the boots on a flat surface and drop the steel ball from a height of 18 inches above the top of the boots in such a manner that the ball strikes the toe area at a point no less than 1.0 inch from the back edge of the metal toe cap. Observe the toe area for fracture of the rubber.

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4.5.6 Leg lining adhesion test. Determine the leg lining adhesion in accordance with Method 5950 of FED-STD-191 except that three 1 by 6-inch specimens shall be taken from the upper leg portion of each of the five sample boots and the rate of travel shall be 2 inches per minute.

4.5.7 Leakage test. Inflate each boot to 0.5 psi to 0.6 psi by use of a boot top plug equipped with an air supply hose, a tube leading to a pressure indicator, and a suitable clamp to retain the plug in the boot top. Immerse the boot vertically in water to an estimated depth of 3-inches and observe for air leaks. Tilt the boot approximately 30 degrees to the left, grasp the heel in one hand and the toe in the other and perform the following operations: Gently flex the sole, twist the boot clockwise and then twist it counter-clockwise, holding each position a sufficient length of time to permit observation for leaks. Repeat the above operation tilting the boot approximately 30 degrees to the right. Submerge the boot to permit observation of the toe area and then the heel area for leaks at the 3-inch level. Return the boot to the vertical position and submerge it to a depth of approximately 9 inches and observe for leaks. Submerge the boot to an estimated depth of 14 inches and observe for leaks. A dwell period of no less than 20 seconds shall be maintained for each leakage observation. The appearance of air bubbles, excluding those considered as surface-trapped air, shall be considered evidence of boot leakage and shall be considered a test failure. NOTE: When type II boot with leg closure is specified, the boot with closure assembly shall be tested for leaks.

4.5.8 Toxicological agents resistance tests (see footnote 1/ to table VI). Two test specimens shall be cut from the area of flexing at the base of the instep on the forward part of each of the ten boots in the sample. Ten specimens, one from each of the boots, shall be tested for mustard vapor penetration as specified in Method T 209 or Method 204 of MIL-STD-282. The other ten specimens shall be tested for GB vapor penetration as specified in Method T 208 or Method 206 of MIL-STD-282.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Type I. Each pair of boots shall be tied together with cotton tape or twine through the metal eyelet.

5.1.1.2 Type II (with ring). Each pair of boots shall be unit packed in a snug-fitting fiberboard box conforming to style FTC, type CF (variety SW), class domestic, grade 175 of PPP-B-636. Containers shall be set up with metal staples, adhesive, or a snap-together tongue or tuck lock corner. Metal staples exposed on the inside of the container shall be covered with pressure-sensitive tape. The boots shall be laid flat inside the container with the

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sole of each boot against the opposite ends of the container. Each container shall be closed in accordance with method I as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.1.2 Commercial. Boots shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing.

5.2.1.1 Type I. Seven pair of boots of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed so that the first pair of boots has the sole of one boot against one end panel and the side of the sole of the second boot resting on the ankle of the first boot within a shipping container. The remaining pairs of boots shall then be placed in the container in the same manner with every other pair reversed. Inside dimensions of each shipping container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Sheets of 30-pound minimum basis weight kraft paper conforming to A-A-203 shall completely separate each boot from the others and shall completely cover the top and bottom of the contents of each shipping container. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.1.2 Type II (with ring). Four pair of boots of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. Level A unit packs shall be packed flat, two in length, one in width, and two in depth within a shipping container. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

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5.2.2 Level B packing.

5.2.2.1 Type I. Seven pair of boots of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A unit packs shall be packed so that the first pair of boots has the sole of one boot against one end panel and the side of the sole of the second boot resting on the ankle of the first boot within a shipping container. The remaining pairs of boots shall then be placed in the container in the same manner with every other pair reversed. Inside dimensions of each shipping container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Sheets of 30-pound minimum basis weight kraft paper conforming to A-A-203 shall completely separate each boot from the others and shall completely cover the top and bottom of the contents of each shipping container. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.2.2 Type II (with ring). Four pairs of boots of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A unit packs shall be packed flat, two in length, one in width, and two in depth within a shipping container. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.2.3 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636, and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.3 Commercial packing. Boots, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), boots packed as specified in 5.2.2 and 5.2.3 shall be palletized on a 4-way entry pallet in accordance with load Type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), Type IV, or Type V in accordance with MIL-STD-147. Pallets shall be

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fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 3 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. Type I boots are intended for use as components of the M5 protective outfit. Type II boots are intended for use as a component of Protective Outfit Toxicological, Microclimate Controlled.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type and sizes required (see 1.2).
c. When a first article is required (see 3.1, 4.3 and 6.3).
d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
e. Type and class of unit load required (see 5.2.1).
f. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.3).
g. When palletization is required (see 5.3).

6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample consisting of one pair of finished boots in any size and width. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.4 Composition. Compounds containing butyl rubber in amounts conforming to table VI have been found to meet the requirements of 3.7.

TABLE VI. Total butyl rubber percentage (by volume)

Compound	Total butyl rubber percentage (minimum)
Upper	63
Lining	66
Friction	61
Outsole	66

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* 6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

6.6 Address for toxicological agents testing (see 4.4.5). The address to which the sample of ten boots for resistance to toxicological agents testing should be forwarded to:

Commander
U.S. Army Armament, Munitions and Chemical Command
ATTN: Product Assurance Directorate, AMSMC-QAO (A)
Test Br./Bldg. E 5100
Aberdeen Proving Ground, MD 21010

6.7 Subject term (key word) listing.

Boots
Chemical protective
Footwear
Toxicological agents protective

6.8 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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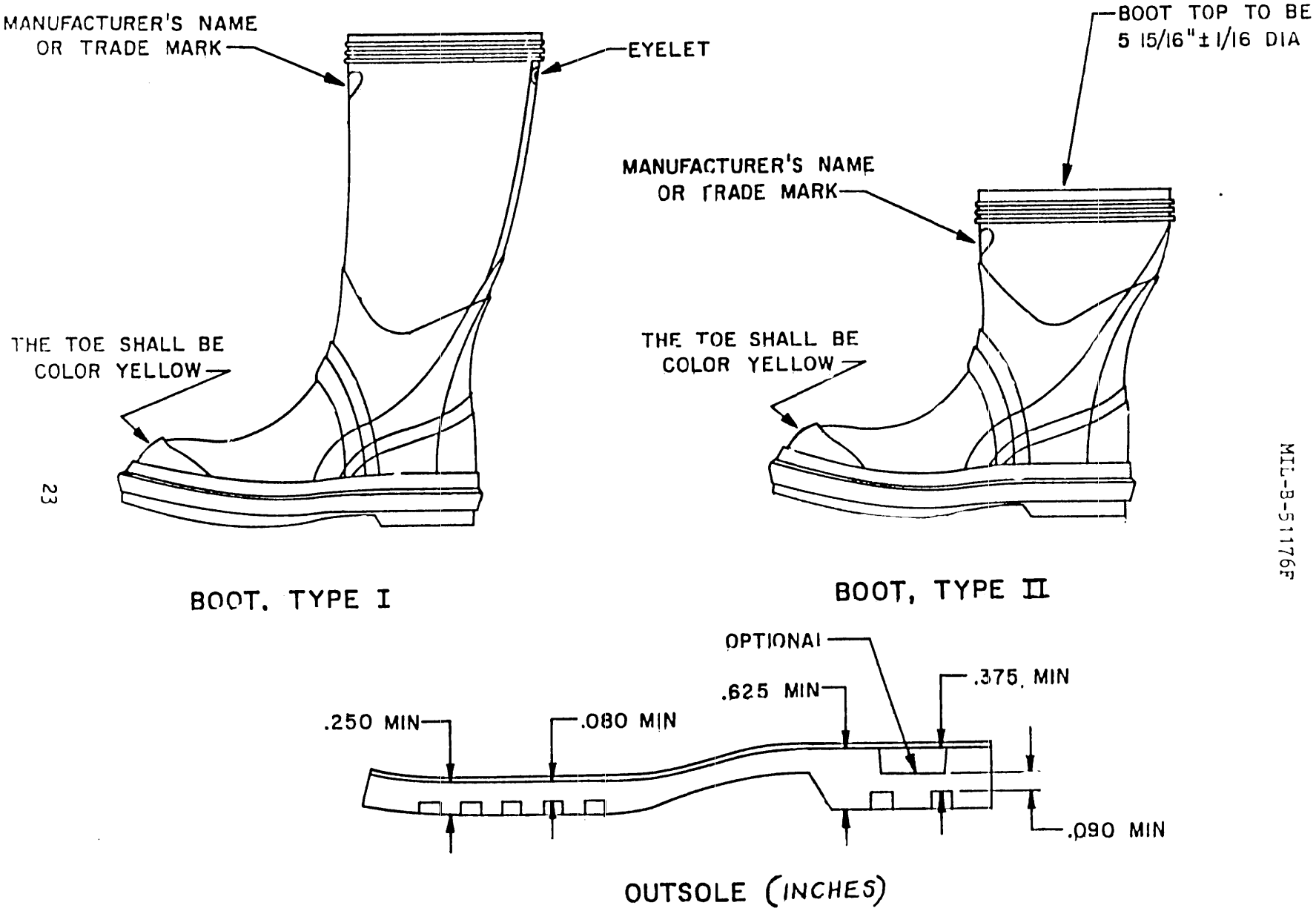


FIGURE 1 BOOTS AND OUTSOLE

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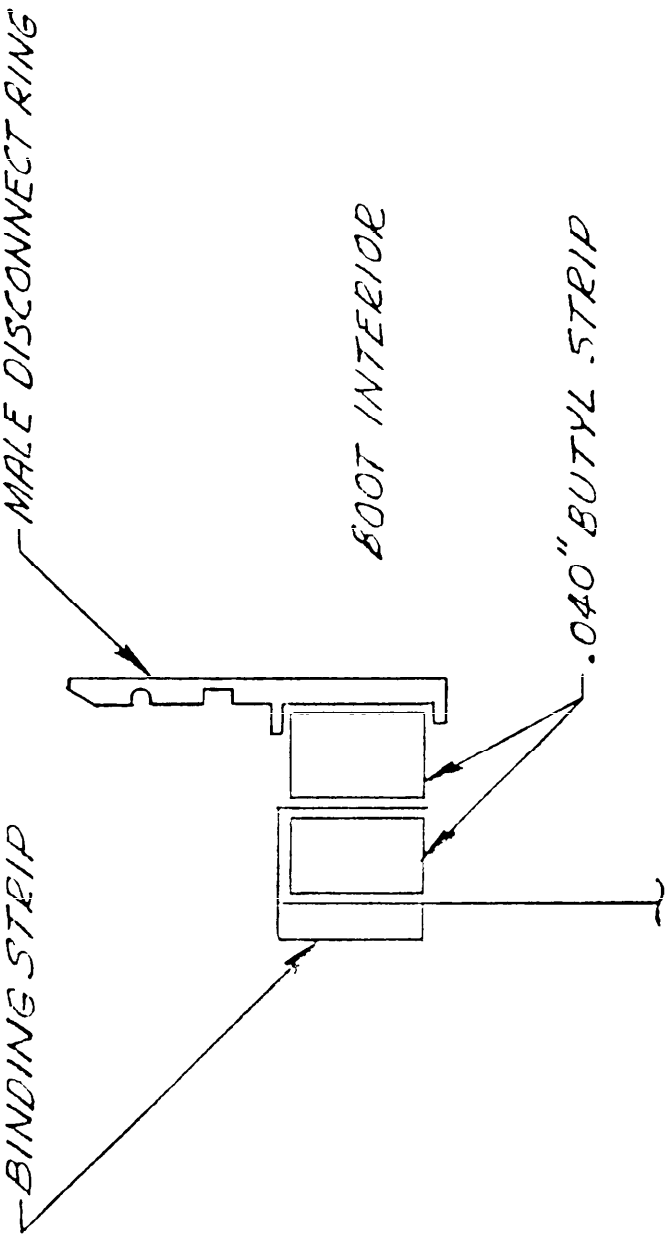


FIGURE 2
ASSEMBLY OF BOOT TO CLOSURE RING
(TYPE II ONLY)

2-1-1499

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-51176F	Boots, Toxicological Agents Protective, M2A1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR
	☐ USER
b. ADDRESS (Street, City, State, ZIP Code)	☐ MANUFACTURER
	☐ OTHER (Specify): _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

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